

CHESAPEAKE BAY BRIDGE TUNNEL, VA, USA

WMO: 997299

Lat: 37.0365N

Lon: 76.0766W

Elev: 16

StdP: 101.13

Time Zone: -5.00 (NAE)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.7	-2.6	N/A	N/A	N/A	N/A	N/A	N/A	17.3	3.4	16.1	3.5	7.4	320	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	31.2	N/A	30.1	N/A	29.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.0	12.8	DB	-6.7	33.9	3.1	1.2	-8.9	34.8	-10.7	35.5	-12.4	36.2	-14.6	37.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	5.1	5.2	8.8	14.6	19.3	24.0	26.1	25.7	23.0	18.1	12.0	9.0
	DBStd	8.32	4.84	4.39	4.47	4.00	3.33	2.70	1.90	1.76	2.39	3.47	3.61	4.57
	HDD10.0	491	166	143	79	7	0	0	0	0	0	1	21	75
	HDD18.3	1760	411	366	297	126	29	1	0	0	1	46	191	292
	CDD10.0	2669	14	11	41	144	288	421	500	486	390	251	81	42
	CDD18.3	895	0	0	1	13	59	172	242	227	141	38	1	1
	CDH23.3	5725	0	0	1	34	178	1094	2016	1729	606	67	0	1
	CDH26.7	1121	0	0	0	1	16	224	483	323	73	1	0	0
Wind	WSAvg	6.3	6.6	6.7	7.1	6.7	6.2	5.7	5.5	5.3	6.2	6.5	6.5	6.2
Precipitation	PrecAvg	1177	84	77	99	80	89	98	138	130	127	95	73	89
	PrecMax	1655	155	246	239	143	141	227	234	287	429	227	252	190
	PrecMin	798	36	30	18	17	27	36	28	31	23	3	3	36
	PrecStd	203	33	45	42	31	28	42	55	72	90	53	54	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.7	19.1	21.8	25.9	28.4	32.3	33.1	32.0	30.5	26.6	21.9	22.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	17.6	16.5	19.3	24.3	26.7	30.4	31.3	30.7	28.8	25.2	20.1	19.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	15.1	14.2	17.6	22.6	25.4	29.1	30.2	29.6	27.2	23.9	18.5	18.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	12.8	11.8	15.8	20.5	24.0	27.8	29.2	28.5	26.2	22.8	17.0	15.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.8	5.8	6.1	5.0	4.4	4.2	3.5	3.2	4.2	4.4	5.3
		MCDBR	8.9	9.5	9.6	8.9	7.1	6.2	6.1	5.5	5.3	5.4	6.6	8.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.310	0.320	0.346	0.392	0.435	0.482	0.511	0.505	0.409	0.355	0.325	0.314	
	taud	2.547	2.512	2.470	2.354	2.265	2.183	2.116	2.120	2.380	2.525	2.558	2.566	
	Ebn at Noon	886	918	920	890	852	809	782	777	842	866	861	857	
	Edh at Noon	77	90	103	122	136	147	156	152	112	88	76	72	
All-Sky Solar Radiation	RadAvg	2.31	3.08	4.14	5.28	5.96	6.41	6.12	5.52	4.50	3.54	2.62	2.06	
	RadStd	0.19	0.32	0.33	0.37	0.51	0.42	0.29	0.32	0.36	0.38	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+105

Nomenclature: See separate page